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Geocoding school addresses

1. Start with the ELSEGIS (Elementary and Secondary Education General Information System) data in 1970.

These steps are done in ***ELSEGIS_school_AEJ.do***.
a. Define elementary and high schools.
Focusing on high schools for now defined as any school with a 12^{th} grade
b. Match OE codes to jurisdiction codes using cross-walk file ***oenum_merge.dta***.
OE codes come at three levels: OEstate; OEdistrict; and OEschool. I string these three together to make OEnum.
c. Produce an output file for geocoding. This file has school OE codes; state, county, zip code, and jurisdiction in which the school is located; school name and school address.

2. School address geocoding (steps done in GIS)

a. Separate the files by county. Output is a .txt file, but rename as a .dbf.

b. Download county-level shapefiles from the ESRI website. Save in a shapefile folder.

c. Open Arcmap. Add two datasets: the county shape file, and the .dbf with the school addresses to geocode. Right-click on the address file and chose "geocode addresses."

d. The program prompts you to add an address locator. The address locators can be made in Arc_catalog. Address locators can use street names only, or can be within zones created by ZIP codes. I used minimum match = 30. Use "us street" as the type. Spelling tolerance = 80

e. Press "OK" and geocoding will begin. For all unmatched schools, eyeball the different possibilities. For remaining addresses that did not match automatically, do the by-hand matching procedure. Go through the interactive match process for any unmatched.
Frequent problems include: misspelled streets; low match scores for various reasons (like no directional); address ranges; missing 'st' or 'av' in intersections. These can be addressed.
Other problems that cannot be addressed: streets that are not found for various reasons, including the fact that from 1970 to 2000 the street name was changed; the address was mis-transcribed on original form, etc.

f. Output is a shapefile with dots for each school that can be laid onto of the tract files (as a layer) in each county. Save the output files. Once each of the files has been saved into the "results" folder, use the Arc_toolbox feature "append" (under data_management/general) to combine all counties into one.

[At this point, turn to the Word file step2_GISmatching.doc to see the rest of the steps in GIS. These steps match Census tract to nearest school within same district.]

3. Match ELSEGIS data with information from the Office of Civil Rights (at the school level, data courtesy of Sarah Reber). The OCR collected information on racial composition at the school level.

Note that 70 schools do not have same oeschool number in two sources. The numbers are not off by 1 or 2 digits; rather, they are *completely* different. The file ***elsegis_ocr_crosswalk.txt*** indicates the ELSEGIS and the OCR numbers for these schools. With this cross-walk, I am able to merge 413 of the 420 high schools in the sample districts.